

Analysis of Correlated Data (연관자료 분석)

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Today's Topics

Analysis of Correlated Data

- Introduction to Correlated Data
- Longitudinal Study
- Models for Correlated Data

Repeated Measurement ANOVA

- Analysis of Variance
- One-way ANOVA
- Fixed vs Random Effects
- Factorial ANOVA
- ANOVA with Repeated Measures

Introduction to Correlated Data

1. Clustered Data

Observations within the same cluster are likely to be correlated

Longitudinal Study

Subject	time					
1	x		x		x	
2			x		x	
...						
m		x	x	x		x

Familial Study

Family	Member		
1	x	x	x
2	x	x	
...			
m	x	x	

Introduction to Correlated Data (Cont'd)

2. Hierarchical Data

Observations from the same hierarchy are likely to be correlated

Survey Sampling

state	county	household				
1	1	x	x	x		
	2	x	x	x	x	
	3	x	x			
2	1	x	x	x	x	x
	2	x	x	x		

Clinical trial

center	physician	patient				
1	1	x	x			
	2	x	x	x	x	
	3	x	x	x		
2	1	x	x	x	x	x
	2	x	x	x		
	3	x	x			
	4	x	x	x		

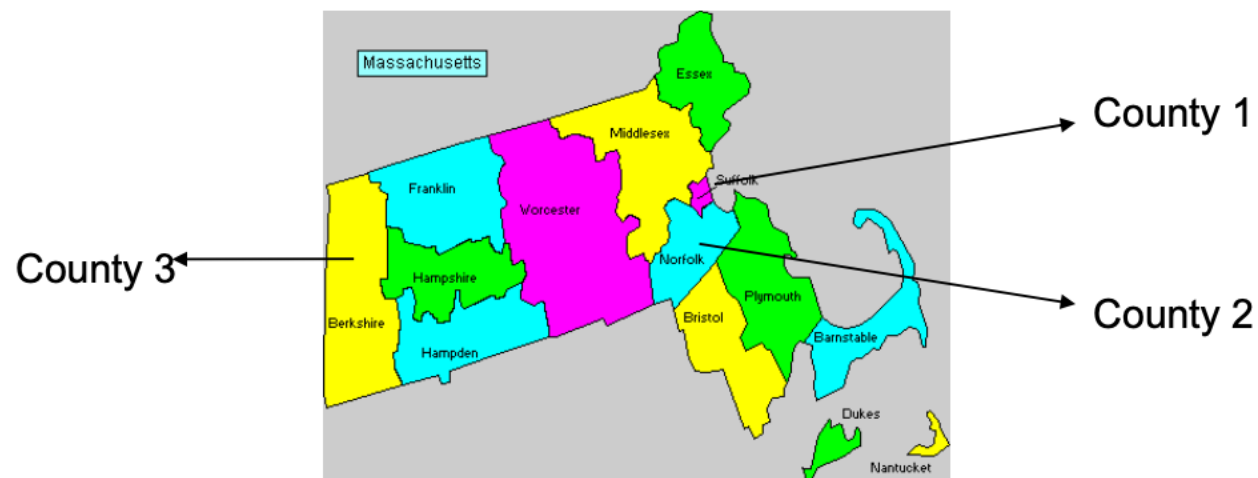
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Introduction to Correlated Data (Cont'd)

3. Spatial Data

Observations are correlated due to spatial proximity

Disease mapping



- The disease rates in counties 1 and 2 are more likely to be similar compared to the disease rates in counties 1 and 3.
- $Y = \# \text{ of disease cases (cases)}/10,000$, $X = \text{exposure}$.

Longitudinal Study

A longitudinal study is a cohort study (follow-up study), in which repeated measurements are taken over time for each individual.

Framingham heart study

- In the Framingham heart study, each of 2634 study participants was examined every two years for 10 years for cholesterol level, BMI, smoking etc.
- The outcome cholesterol is a continuous variable.

	0	2	4	6	8	10 (year)
cholesterol	X	X	X	X	X	X
BMI	X	X	X	X	X	X
smoking	X					
age	X					

- Study objectives

How does cholesterol level change over time?

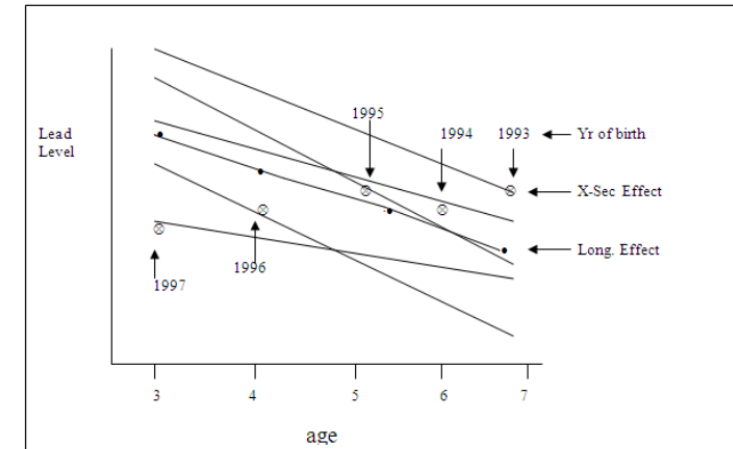
How is cholesterol level associated with BMI, smoking and age?

Longitudinal Study (Cont'd)

Data structure

	Cross-sectional Study		Longitudinal Study	
	Subject	Data	Subject	Data
X=BMI Y=CHOL	1	X_1	1	$X_{11} X_{12} \dots X_{15}$
		Y_1		$Y_{11} Y_{12} \dots Y_{15}$
	2	X_2	2	$X_{21} X_{22} \dots X_{25}$
		Y_2		$Y_{21} Y_{22} \dots Y_{25}$
				

Figure: Longitudinal vs. Cross-Sectional Effects in Blood Lead Level Data



- Classical linear/logistic models for cross-sectional data has a single measure of X and Y for each subject, while longitudinal studies have multiple measures of X and Y for each subject.
- A longitudinal study allows us to estimate the between-subject variation and the within-subject variation.
- It is robust to confounding effect.

Longitudinal Study (Cont'd)

In classical linear/logistic regression, the key assumption is the independence between observations.

For example, $Y = \text{SBP}$, $X_1 = \text{age}$, $X_2 = \text{sex}$

1st subject's SBP does not depend on 2nd subject's SBP

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \varepsilon_i$$

However, in a longitudinal study, the observations over time from the same subject are likely to be correlated.

		time		
		1	2	3
SBP	Subject 1	150	142	157
	Subject 2	100	105	97

- Subject 1 is independent of subject 2.
- Observations for the same subject are correlated.

A valid statistical inference must take the within-subject correlation into account.

Models for Correlated Data

1. Generalized Linear Mixed Model (GLMM)

Linear mixed model: Laird and Ware (1982, *Biometrics*)

Generalized linear mixed model: Breslow and Clayton (1993, *JASA*)

Model: $g(\mu_{ij}) = x_{ij}^T \beta + z_{ij}^T b_i$, where $g()$ is a link function and $b \sim N(0, D(\theta))$

Features:

- Mixed effect model (Subject-specific model).
- Model correlation explicitly using random effects in the modeling stage.
- Likelihood inference is straightforward in principle.
- Difficulty: likelihood is straightforward in principle, but often involves numerical integration
- Inference requires a correct specification of the correlation structure.

Models for Correlated Data (Cont'd)

2. Generalized Estimating Equations (GEEs)

Zeger, Linag and Self (1985, *Biometrika*)

Liang and Zeger (1986, *Biometrika*)

Extension of GLM using quasi-likelihood estimation

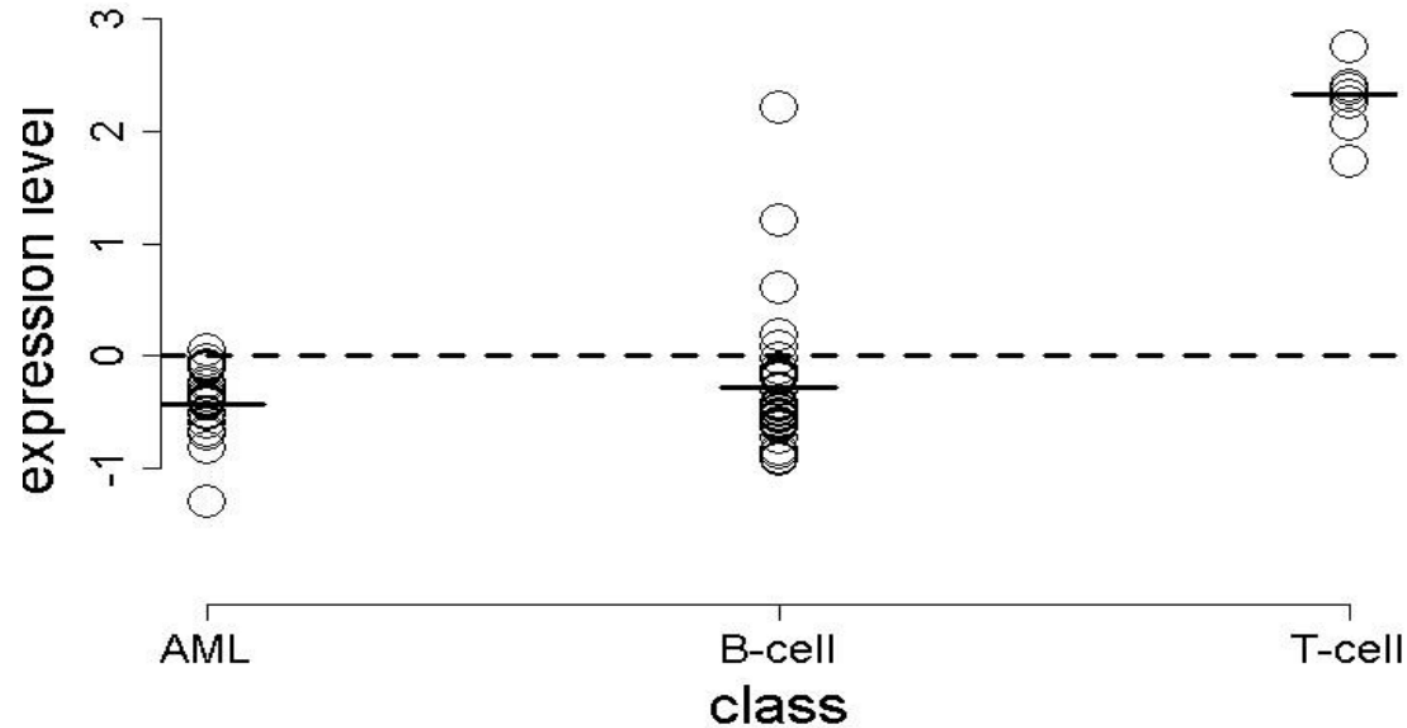
Model: $g(\mu_{ij}) = x_{ij}^T \beta$

Features:

- Marginal Model (Population-Average Model), i.e. modeling the population mean. That is, it avoids a need for multivariate distributions by only assuming a functional form for the marginal distribution at each timepoint.
- Inference is robust to misspecification of the correlation structure and the sandwich estimator is used to correct for misspecification of the correlation structure.
- No likelihood. Likelihood-based inference (e.g., LR test) is difficult.

Analysis of Variance

Main Idea: Evaluate the effect of treatment by analyzing the amount of variation among the subgroup sample means



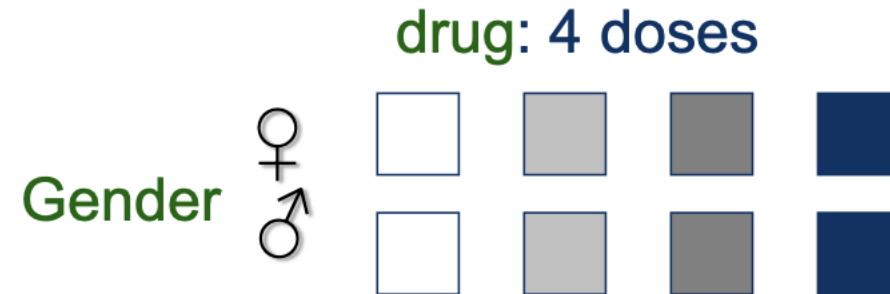
Analysis of Variance (Cont'd)

ANOVA tests mean differences between groups defined by categorical variables

One-way ANOVA: one factor with 2 or more levels



Multi-way ANOVA: 2 or more factors with 2 or more levels



Assumptions

- Independence of cases: this is a requirement of the design.
- Normality: the distributions in each cells are normal.
- Homogeneity of variances - the variance of data in groups should be the same.

Analysis of Variance (Cont'd)

Nonnormality

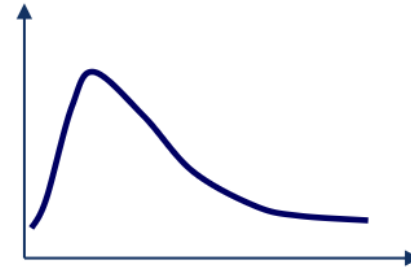
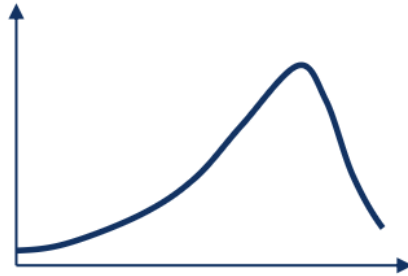
- F-test is very robust against non-normal data, especially in a fixed-effects model.
- Large sample size will approximate normality by Central Limit Theorem (recommended sample size > 50)
- A large deviation from normality leads to hypothesis test conclusions that are too liberal

Remedy for nonnormality

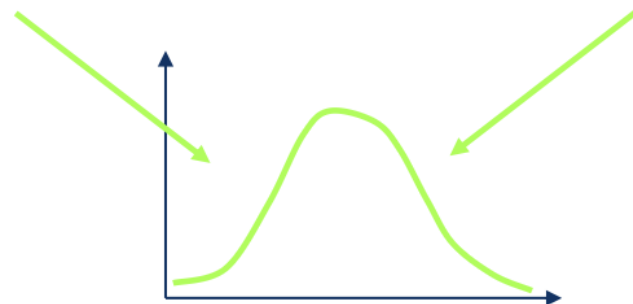
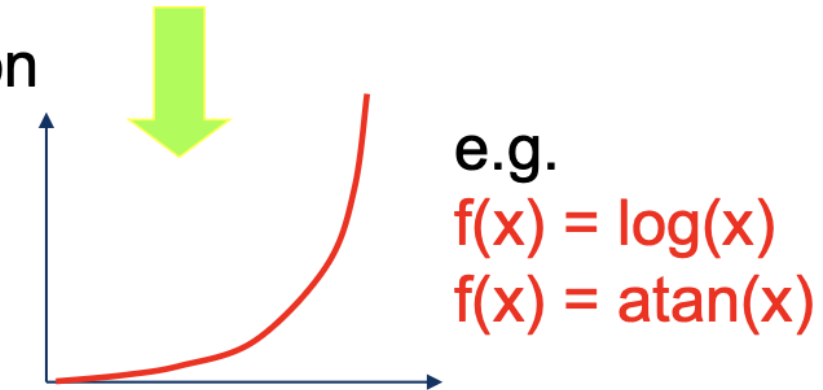
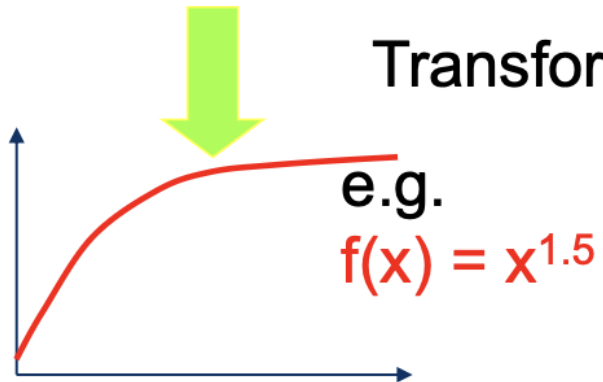
- Data transformation
- Modified F-tests
- Permutation test on the F-statistics

Analysis of Variance (Cont'd)

Transformation



Transformation



Analysis of Variance (Cont'd)

Permutation

- Distribution of statistics is empirically estimated with the permuted samples under the null hypothesis.
- P-value can be obtained under the null hypothesis.
- It can be applied to any statistics.

(Example)

Group 1: 10, 15, 20 vs Group 2: 3, 4, 5

We are interested in getting p-values for $H_0: \mu_1 = \mu_2$

Number of possible permuted sample: $\binom{6}{3} = 20$

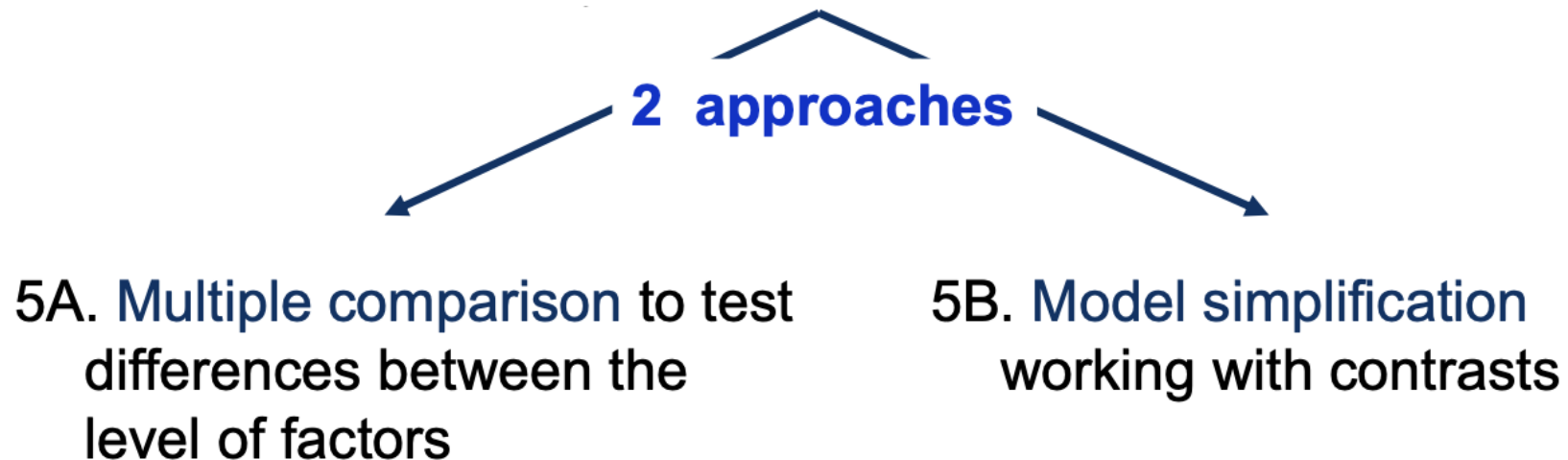
Analysis of Variance (Cont'd)

Permutation

Label	Group I			Group II			test statistics
1	3	4	5	10	15	20	-3.737
2	3	4	10	5	15	20	-1.558
3	3	5	10	4	15	20	-1.356
4	4	5	10	3	15	20	-1.178
5	3	4	15	5	10	20	-0.741
6	3	5	15	4	10	20	-0.615
7	4	5	15	3	10	20	-0.495
8	3	4	20	5	10	15	-0.161
9	3	5	20	4	10	15	-0.053
10	3	10	15	4	5	20	-0.053
11	4	5	20	3	10	15	0.053
12	4	10	15	3	5	20	0.053
13	5	10	15	3	4	20	0.161
14	3	10	20	4	5	15	0.495
15	4	10	20	3	5	15	0.615
16	5	10	20	3	4	15	0.741
17	3	15	20	4	5	10	1.178
18	4	15	20	3	5	10	1.356
19	5	15	20	3	4	10	1.558
20	10	15	20	3	4	5	3.737

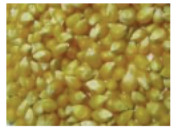
One-way ANOVA

1. Test normality
2. Test homogeneity of variance within each group
3. Run the ANOVA
4. Reject/accept the H_o (all the means are equal)

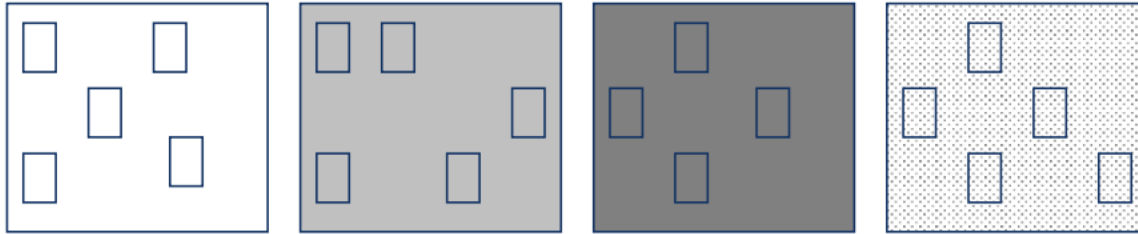


One-way ANOVA (Cont'd)

(Example)



Maize: 4 varieties (k)



y: productivity (NORMAL CONTINUOUS)
x: variety (CATEGORICAL: four levels: x_1, x_2, x_3, x_4)

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_a : At least two means differ

ANOVA model

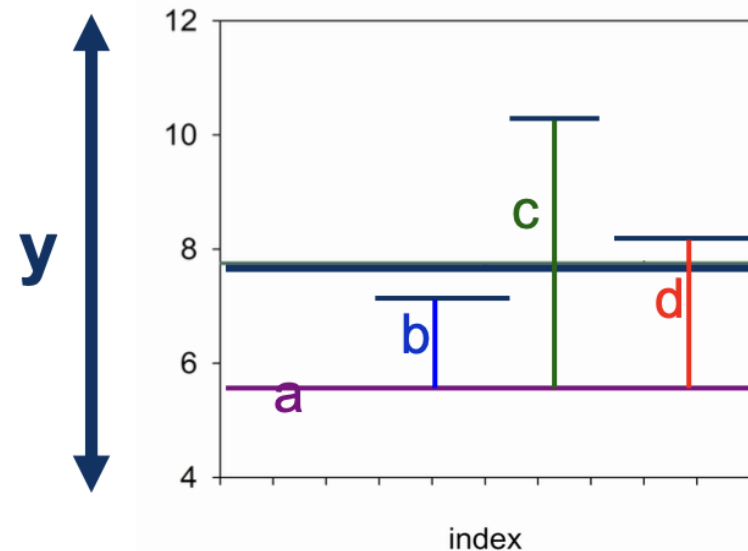
$$y_i = a + bx_2 + cx_3 + dx_4$$

$$a = \mu_1$$

$$c = \mu_1 - \mu_3$$

$$b = \mu_1 - \mu_2$$

$$d = \mu_1 - \mu_4$$



One-way ANOVA (Cont'd)

(Example)

	Var 1	Var 2	Var 3	Var 4
	6.08	6.87	10.26	8.79
	5.7	6.77	10.21	8.42
	6.5	7.4	10.02	8.31
	5.86	6.63	9.65	8.57
	6.17	6.98	X	9.03
n_i	5	5	4	5
μ_i	6.06	6.93	10.03	8.62

Number of groups: $k = 4$

Number of observations $N = 19$

Grand mean = 7.80

Sum of squares (SS): deviance

$$SS \text{ Total} = \sum (y_i - \text{grand mean})^2$$

$$SS \text{ Factor} = \sum n_i (\text{group mean}_i - \text{grand mean})^2$$

$$SS \text{ Error (within group)} = \sum (y_i - \text{group mean}_i)^2$$

Degree of freedom (df)

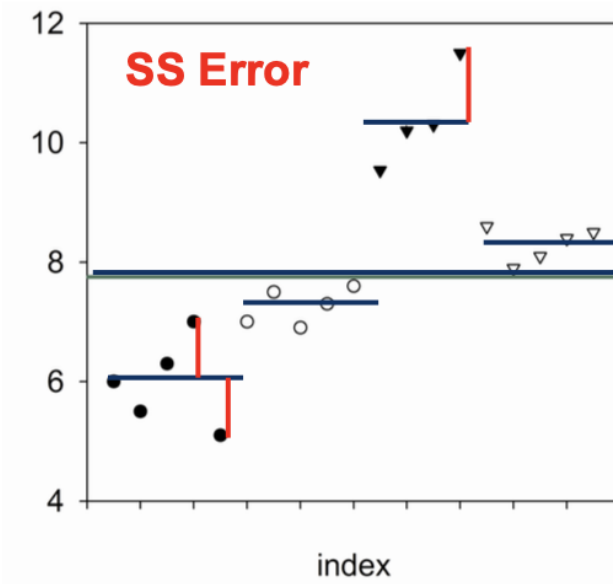
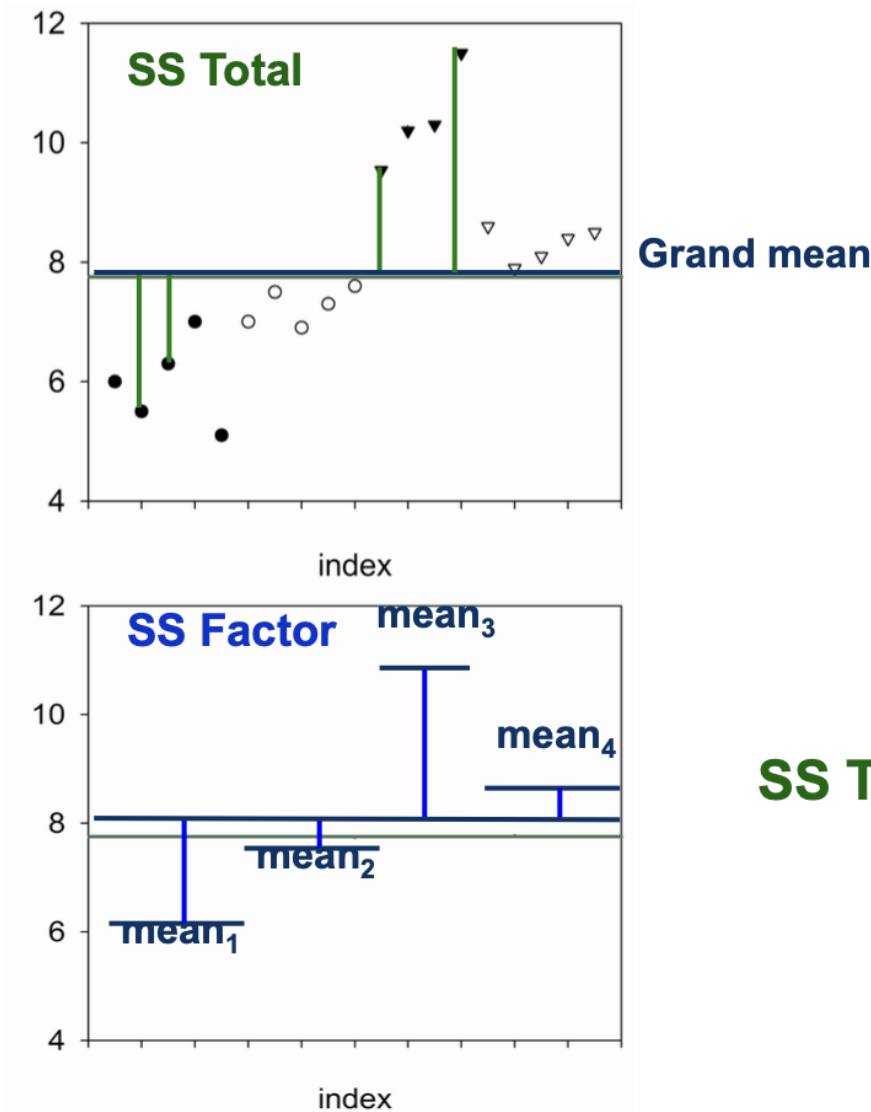
$$\text{Total: } N - 1$$

$$\text{Group: } k - 1$$

$$\text{Error: } N - k$$

One-way ANOVA (Cont'd)

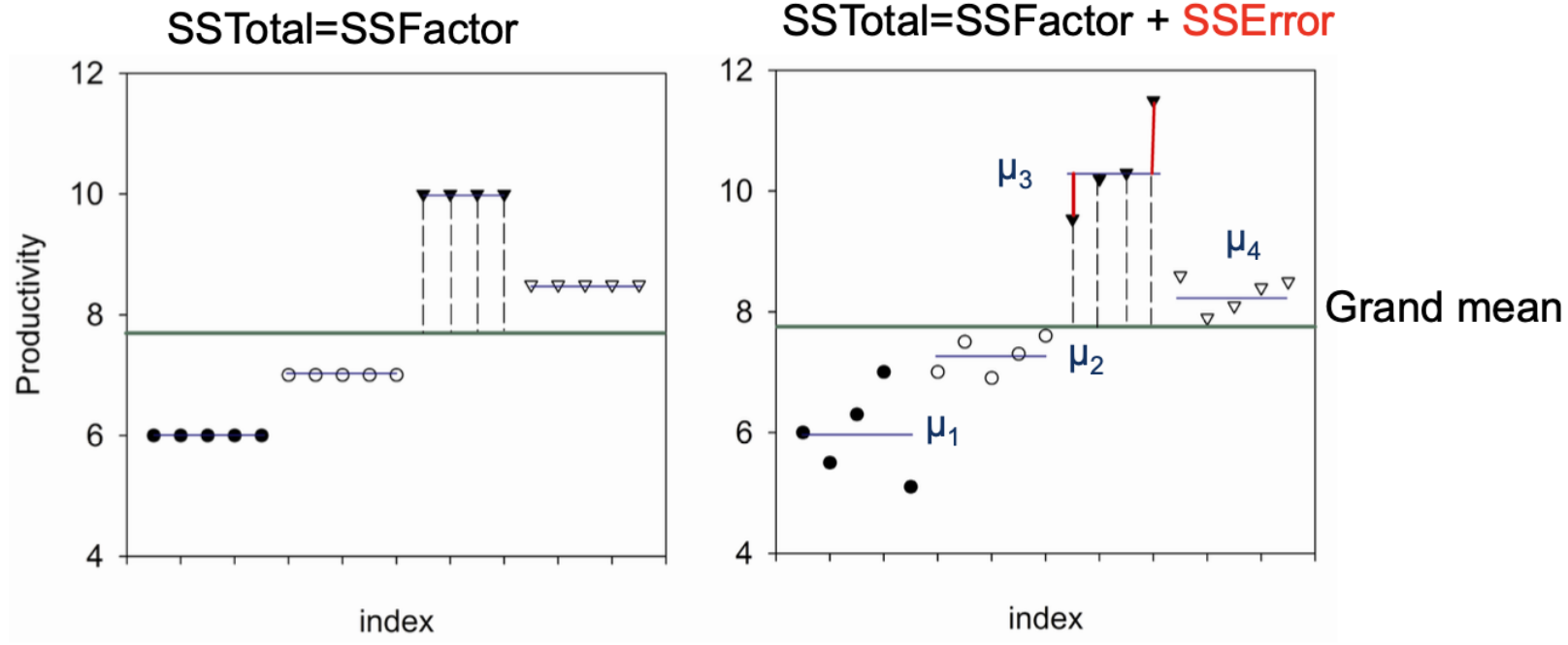
(Example)



SS Total = SS Factor + SS Error

One-way ANOVA (Cont'd)

(Example)



SS can be divided by the respective df to get a variance

$MS = SS/df$ (Mean squared deviation)

One-way ANOVA (Cont'd)

If the k means are not equal, then the Factor MS in the population will be greater than the population's Error MS.

$$F = \frac{\text{Factor MS}}{\text{Error MS}}$$

How to define the correct F statistic can be a difficult task with complex design.

Multi-way ANOVA

Multi-way ANOVA is used when the experimenter wants to study the effects of two or more treatment variables.

Assumptions

- Independence of cases: this is a requirement of the design.
- Normality: the distributions in each cells are normal.
- Homogeneity of variances - the variance of data in groups should be the same.
- Equal replication - balanced and orthogonal design

	Dose 1	Dose 2	Dose 3
Low temp	- X	10 obs	10 obs
High temp	10 obs	10 obs	8 obs X

Fixed vs Random Effects

If we consider more than one factor we have to distinguish two kinds of effects

Fixed effects: factors are specifically chosen and under control, they are informative.

Random effects: factors are chosen randomly within a large population, they are normally not informative.

Random effects mainly occur in two contrasting kinds of circumstances

1. Observational studies with hierarchical structure
2. Designed experiments with different spatial or temporal dependence

Fixed vs Random Effects (Cont'd)

Imagine that we randomly select a of the possible levels of the factor of interest. In this case, we say that the factor is random if their effects can be assumed to follow the normal distribution.

Typically random factors are categorical.

While continuous covariates may be measured at random levels, we usually think of the effects as being systematic (such as linear, quadratic or even exponential) effects. Random effects are not systematic.

Why is it so important to identify fixed vs random effects?

- They affect the way to construct the F-test in a multifactorial ANOVA.

- Their misspecification leads to wrong conclusions

If we have only random effect, then we are working on Random Effect model.

If we have both fixed and random effects, then we are working on Mixed model.

Fixed vs Random Effects (Cont'd)

(Example) Four looms have been chosen randomly from a population of looms within a weaving shed and four observations of fabric strength were made on each loom. The data obtained from the experiment are below.

Loom	Obs 1	Obs 2	Obs 3	Obs 4	row sum
1	98	97	99	96	390
2	91	90	93	92	366
3	96	95	97	95	383
4	95	96	99	98	388

$$y_{ij} = \mu + r_i \text{ (random)} + \varepsilon_{ij}, r_i \sim N(0, \sigma_1^2), \varepsilon_{ij} \sim N(0, \sigma_2^2)$$

$$y_{ij} = \mu + \alpha_i + r_i + \varepsilon_{ij}, r_i \sim N(0, \sigma_1^2), \varepsilon_{ij} \sim N(0, \sigma_2^2)$$

Factorial ANOVA: two or more factors

Factorial design: two or more factors are crossed. Each combination of the factors equally replicated and each factor occurs in combination with every level of the other factors

4 fertilizer	3 levels of irrigation		
	10	10	10
	10	10	10
	10	10	10
	10	10	10

Factorial ANOVA: two or more factors (Cont'd)

Why use a factorial ANOVA? Why not just use multiple one-way ANOVAs?

With n factors, you'd need to run n one-way ANOVA's, which would inflate your α -level. However, this could be corrected with a Bonferroni correction

The best reason is that a factorial ANOVA can detect interactions, something that multiple one-way ANOVA's cannot do.

It should be noted that orthogonality is not often preserved in observational study.

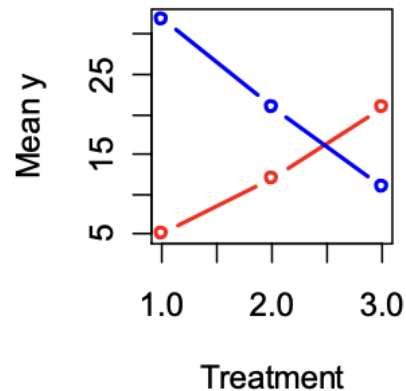
If they are not orthogonal, true effects of factor A can be estimated only if both A and B are included as covariates. Furthermore if A and B are highly correlated, their effects are not distinguishable because of multi-collinearity.

Factorial ANOVA: two or more factors (Cont'd)

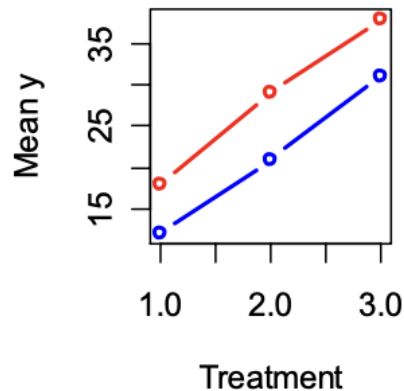
Interaction: When the effects of one independent variable differ according to levels of another independent variable

Main effects: the effect of a factor is independent from any other factors

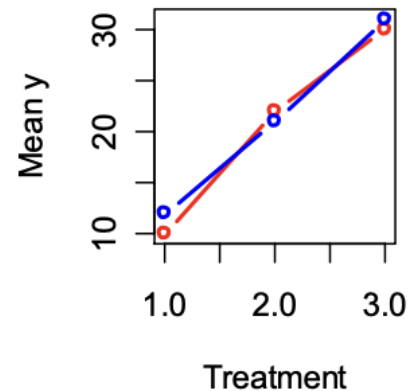
Effect	
Clone	< 0.05
Treat	< 0.05
C x T	< 0.05



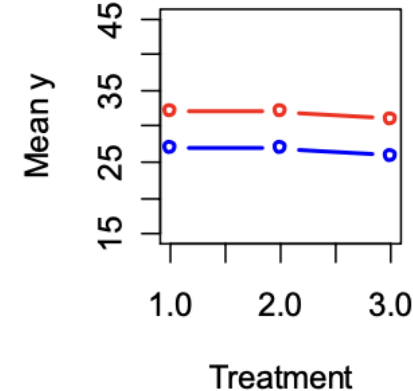
Effect	
Clone	< 0.05
Treat	< 0.05
C x T	n.s.



Effect	
Clone	n.s.
Treat	< 0.05
C x T	n.s.



Effect	
Clone	< 0.05
Treat	n.s.
C x T	n.s.



Two-crossed factor design

CloneA —
CloneB —

Treatment: 3 levels

Factorial ANOVA: two or more factors (Cont'd)

Two crossed fixed effects: every level of each factor occurs in combination with every level of the other factors

Model 1: two fixed effects

Model 2: two random effects (uncommon situation)

Model 3: one random and one fixed effect

We can test main effects and interaction:

The main effect of each factor is the effect of each factor independent of (pooling over) the other factors

The interaction between factors is a measure of how the effects of one factor depends on the levels of one or more additional factors

We can only measure interaction effects in factorial (crossed) designs.

Factorial ANOVA: two or more factors (Cont'd)

(Example) Two crossed fixed-effects

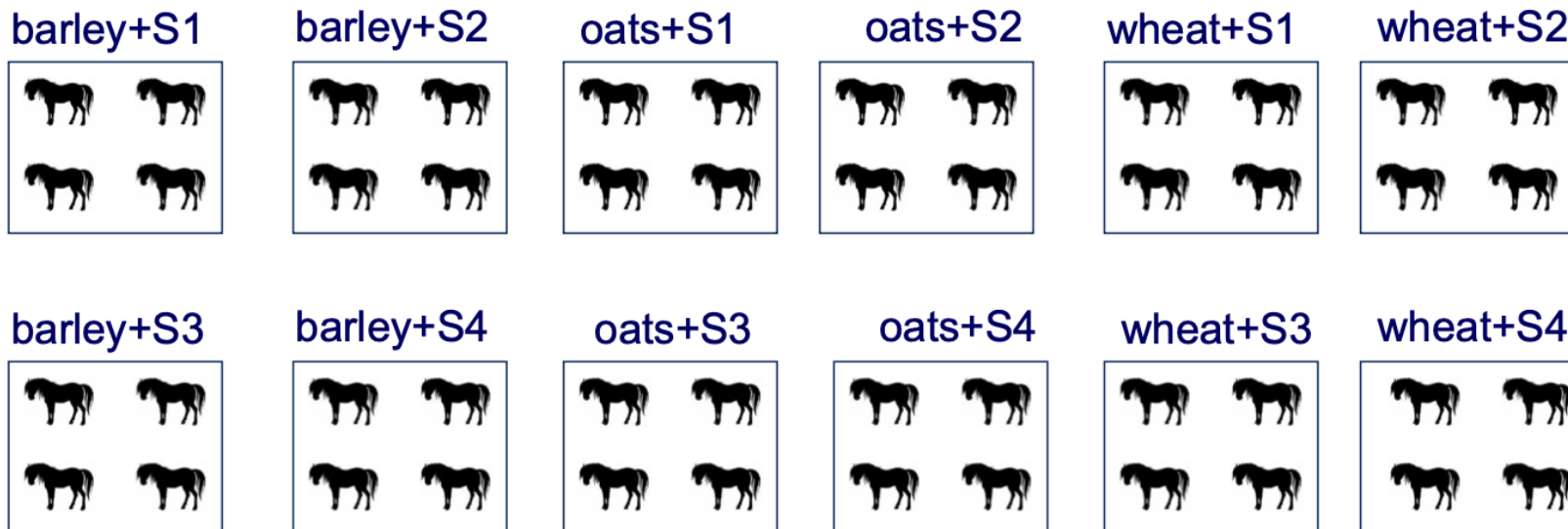
Response variable: weight gain in six weeks

Factor A: DIET (3 levels: barley, oats, wheat)

Factor B: SUPPLEMENT (4 levels: S1, S2, S3, S4)

DIET* SUPPLEMENT = $3 \times 4 = 12$ combinations

We have 48 horses to test our two factors: 4 replicates



Factorial ANOVA: two or more factors (Cont'd)

(Example) Two crossed fixed-effects

F test for main effects and interaction:

$$\text{DIET} \quad F_D = \frac{MS_D}{MS_{error}}$$

$$\text{SUPPLEMENT} \quad F_S = \frac{MS_S}{MS_{error}}$$

$$\text{DIET*SUPPLEMENT} \quad F_{D \times S} = \frac{MS_{D \times S}}{MS_{error}}$$

Factorial ANOVA: Mixed Models

Pure fixed effect models require independence

What if we have dependence?

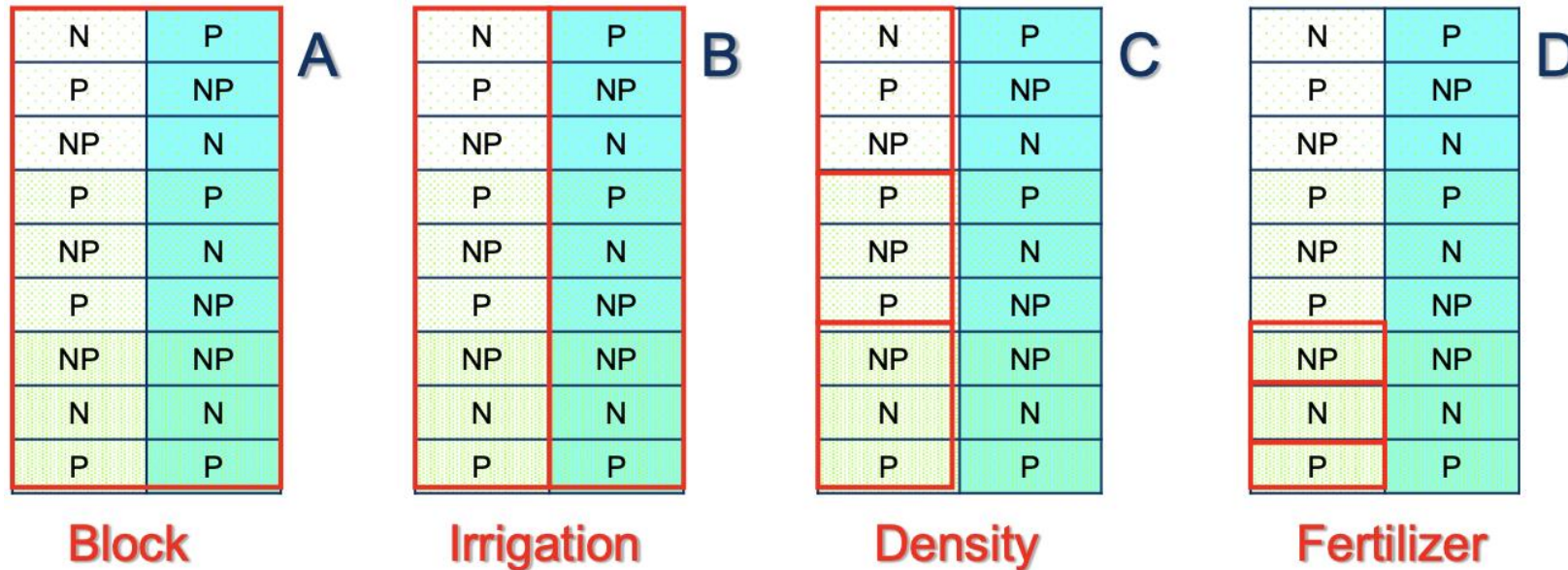
Mixed models can deal with spatial or temporal dependence.

The mixed models included the dependence in the data with appropriate random effects

Factorial ANOVA: Mixed Models (Cont'd)

The split-plot design is one of the most useful design

The different treatments are applied to plots with different sizes organized in a hierarchical structure



Response variable: Crop production in each cell

Factorial ANOVA: Mixed Models (Cont'd)

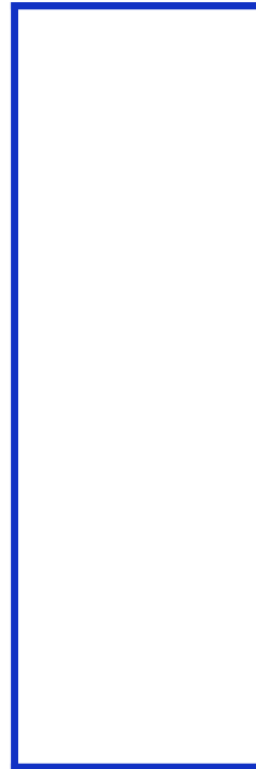
The split-plot design

No water	Water
N	P
P	NP
NP	N
P	P
NP	N
P	NP
NP	NP
N	N
P	P

i) 4 Blocks



ii) irrigation plot (yes or no)



**HIERARCHICAL
STRUCTURE**



iii) seed-sowing
density sub-plots
(low, med, high)



iv) 3 fertilizer
sub-sub-plot
(N, P, o NP)



Crop production

Factorial ANOVA: Mixed Models (Cont'd)

Model formulation

$Y \sim \text{fixed effects} + \text{error terms}$

$y \sim a*b*c + \text{Error}(a/b/c)$



Here you can
specify your
sampling hierarchy

Yield



Block

Irrigation
Density
Fertilizer

Uninformative

Informative!!!

Factorial ANOVA: Mixed Models (Cont'd)

Mixed models using traditional ANOVA requires perfect orthogonal and balanced design.

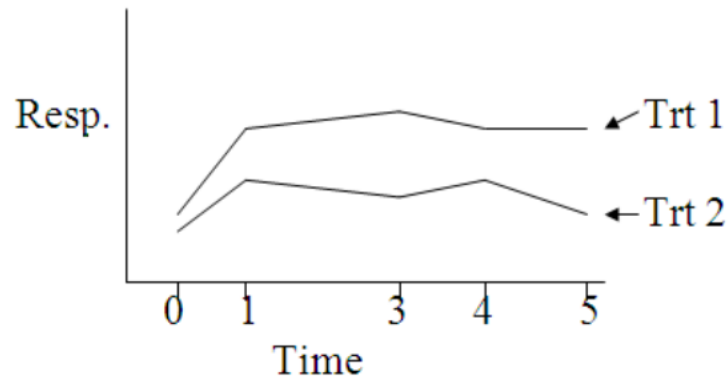
It should be avoided to work with F-test for multi-way ANOVA in non-orthogonal sampling designs.

If something has gone wrong with the sampling, you can use linear mixed models with the REML estimation.

ANOVA with Repeated Measures

Statistical model

Plot of means by time for each of two treatments



Sub	1	...	t
1	y_{11}	...	y_{1t}
$\vdots$	$\vdots$		$\vdots$
n	y_{n1}	...	y_{nt}

A standard ANOVA model

Model $y_{hij} = \mu_{hj} + \varepsilon_{hij}$

μ_{hj} the expected response of a subject on treatment h at time t_j ,

ε_{hij} the independent errors, $N(0, \sigma^2)$.

$h = 1, \dots, g$ (# of groups);

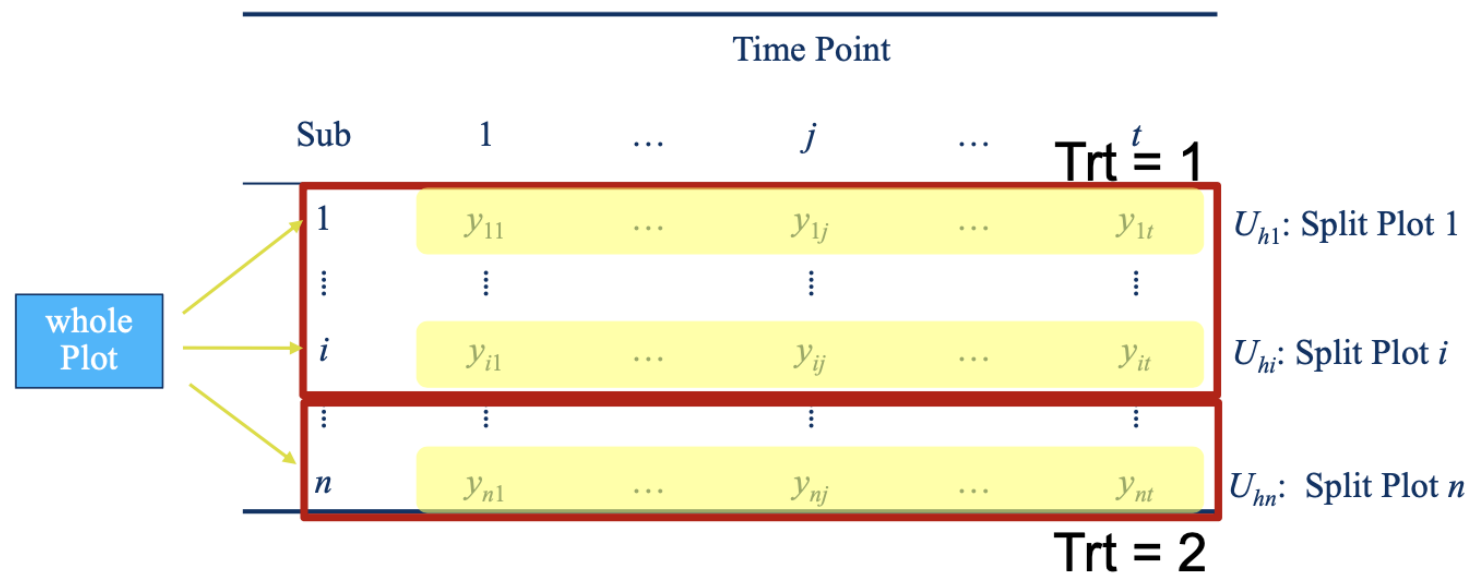
$i = 1, \dots, n_h$ (# of subjects in the h th group);

$j = 1, \dots, k$ (# of measurements).

ANOVA with Repeated Measures (Cont'd)

Mixed effects model

The random effect for an individual is a random displacement term that applies to all measurements for that individual.



Subjects are not nested for treatment

ANOVA with Repeated Measures (Cont'd)

Mixed effects model

The random/mixed effects model is alternately referred to a split plot, hierarchical, two-stage, or variance components model, or the random intercept model.

$$y_{hij} = \mu_{hj} + U_{hi} + \varepsilon_{hij}$$

μ_{hj} the expected response of a subject on trt h at time t_j

U_{hi} the unobserved random effect for subject i in group h ;

$$h = 1, \dots, g; j = 1, \dots, k; i = 1, \dots, n_h;$$

$$U_{hi} \sim N(0, \sigma^2) \text{ for all } h, i ;$$

$$\varepsilon_{hij} \sim N(0, \sigma^2) \text{ are residual errors;}$$

$$U_{hi}, \varepsilon_{hij} \text{ are independent}$$

ANOVA with Repeated Measures (Cont'd)

Mixed effects model

Decomposition of variance of covariance

$$\text{Var}(y_{hij}) = v^2 + \sigma^2$$

v^2 : the population variance of effect

σ^2 : the measurement variance

This assumes the equal covariance between y_{hij} and $y_{hij'}$. However it is not always true and should be confirmed with statistical inference.

Hierarchical or two-stage model is applicable to the mixed-effects model.

$$1. E(y_{hij} | U_{hi}) = \mu_{hj} + U_{hi} = U_{hi}^*, \quad V(y_{hij} | U_{hi}) = \sigma^2$$

$$2. U_{hi} \sim N(0, v^2)$$

The covariance/correlation between measurements within a subject?

ANOVA with Repeated Measures (Cont'd)

Mixed effects model

Why not use 'subject' as a fixed effects?

May require a lot of additional parameters (depending on number of subjects).
Their effects are small and can be assumed to follow the normal distribution.
Implies interest in estimating a separate effect for each person.

Comments

Model fitting does not require estimation of subject effects.

Instead of using a separate parameter (fixed effect) for each subject, we assume subject effects are random realization from a population.

The only additional parameter is the random effect variance, ν^2

ANOVA with Repeated Measures (Cont'd)

Repeated measure ANOVA

It represents an univariate approach to analyzing repeated measures data with particular focus on hypothesis testing.

It can be thought of as an extension of the paired-sample t -test to include comparison between *more than two repeated*.

ANOVA with Repeated Measures (Cont'd)

Between-subject and within-subject effects

Between-subject effects

- Each subject is exposed to one of the treatments being tested.
- Estimate $E(Y \mid E = 1) - E(Y \mid E = 0)$.
- Example: one-way ANOVA.

Within-subject effects

- Each subject is exposed to more than one of the treatments being tested.
- Estimate $E(Y_{i|E=1} - Y_{i|E=0})$, i : subject
- Example: Paired t-test.

Mixed (within and between-subjects) design

- There is at least one within-subjects factor and at least one between-subjects factor in the same experiment.

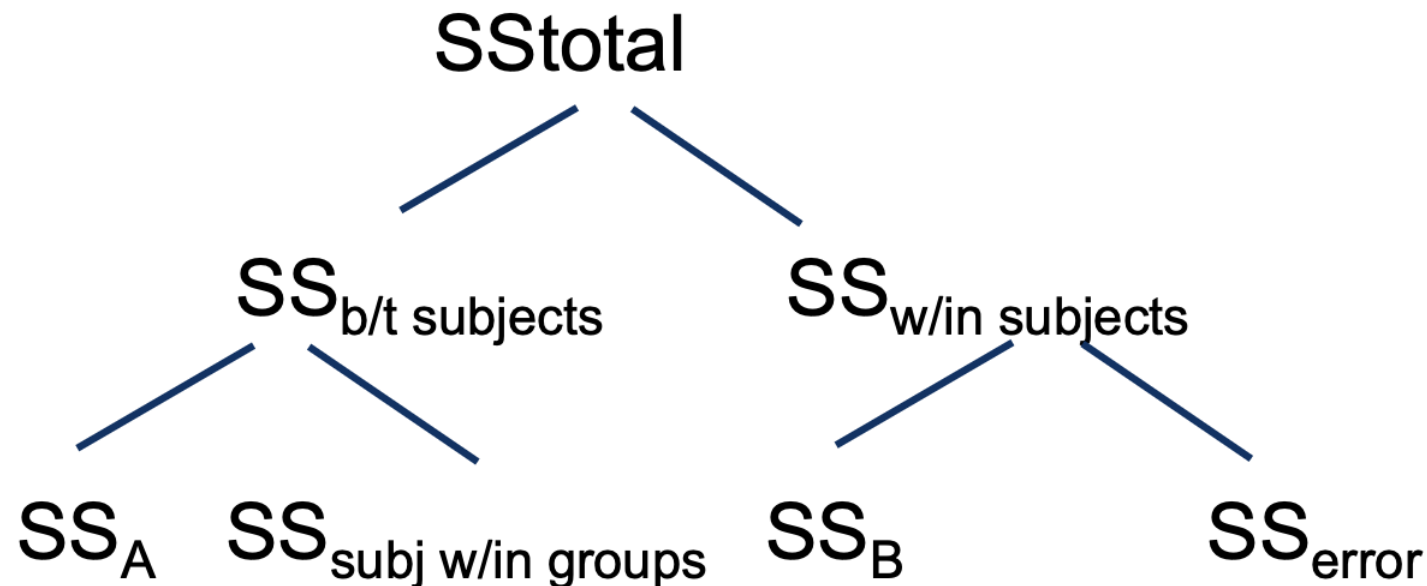
ANOVA with Repeated Measures (Cont'd)

Partition of SS

A x (B x S): A - between, B - within, S - subjects

Each level of factor A contains a different group of (for example) randomly assigned subjects.

On the other hand, each level of factor B at any given level of factor A contains the same subjects.



ANOVA with Repeated Measures (Cont'd)

Partition of SS

SS_{total} : Deviation of each individual score from the grand mean

$SS_{b/t\ subjects}$:

- Deviation of subjects' individual means (across treatments) from the grand mean.
- This is largely uninteresting, as we can pretty much assume that 'subjects differ'.

$SS_{w/in\ subjects}$ - How SS vary about their own mean, breaks down into:

SS_B : As in between subjects ANOVA, it is the comparison of treatment means to each other (by examining their deviations from the grand mean).

SS_{error} : Variability of individuals' scores about their treatment mean.

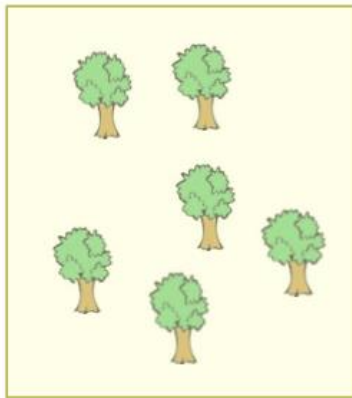
ANOVA with Repeated Measures (Cont'd)

(Example) The Sitka spruce 1989

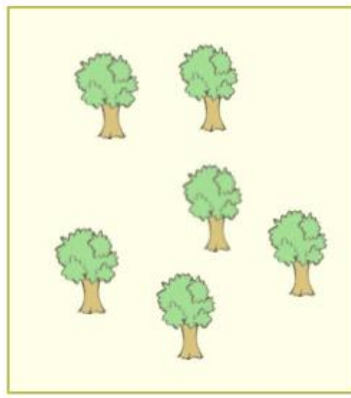
Goal: to investigate the effects of ozone on tree growth.

79 trees are grown in four controlled environment chambers.

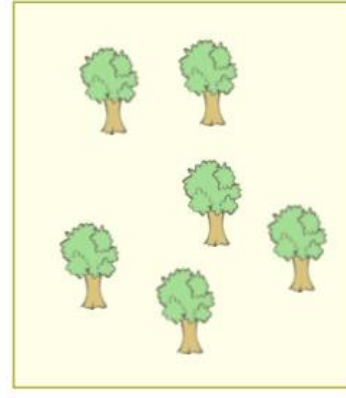
Observations: 103, 130, 162, 190, 213, 247, 273, 308 (days)



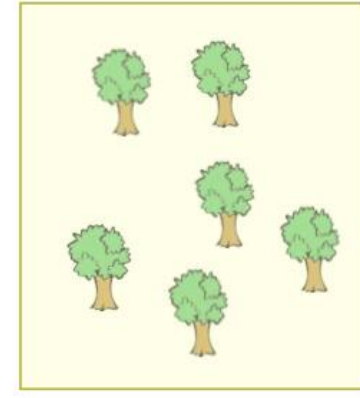
27 trees,
ozone 70 ppb



27 trees,
ozone 70 ppb



12 trees,
No ozone



12 trees,
No ozone

ANOVA with Repeated Measures (Cont'd)

(Example) The Sitka spruce 1989

Model

$$y_{hij} = \mu + \alpha_h + \tau_j + \lambda_{hj} + U_{hi} + \varepsilon_{hij}$$

y_{hij} the j th measurement (response) ($j=1, \dots, n$) for the i th subject ($i=1, \dots, m_h$) in group h ($h=1, \dots, g$);

μ the unknown intercept;

α_h the effect of treatment h , $\sum_h \alpha_h = 0$;

τ_j the effect of time j , $\sum_j \tau_j = 0$ for all h ;

λ_{hj} the effect of treatment h at time j ;

$\sum_j \lambda_{hj} = 0$ for all h , $\sum_h \lambda_{hj} = 0$ for $j=1, \dots, n$;

U_{hi} the random effect of the i th subject in the h th group, $U_{hi} \sim IN(0, v^2)$;

$\varepsilon_{hij} \sim IN(0, \sigma^2)$ are residual errors;

U_{hi} , ε_{hij} are independent.

m $\sum_h m_h$

F_1 : To test for difference in expected response between treatments.

$$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_g \text{ vs } H_1: \text{not } H_0.$$

F_2 : To test for lack of parallel group mean profile.

$$H_0: \lambda_{11} = \lambda_{12} = \dots = \lambda_{gn} \text{ vs } H_1: \text{not } H_0.$$

ANOVA with Repeated Measures (Cont'd)

(Example) The Sitka spruce 1989

ANOVA table

Source of variation	Sum of squares	df
Between treatments	$BTSS_1 = n \sum_{h=1}^g m_h (y_{h..} - y_{...})^2$	$g - 1$
Whole plot residual	$RSS_1 = TSS_1 - BTSS_1$	$m - g$
Whole plot total	$TSS_1 = n \sum_{h=1}^g \sum_{i=1}^{m_h} (y_{hi.} - y_{...})^2$	whole plot
Between times	$BTSS_2 = m \sum_{j=1}^n (y_{..j} - y_{...})^2$	split plot $n - 1$
Treatment by time interaction	$ISS_2 = \sum_{j=1}^n \sum_{h=1}^g m_h (y_{h.j} - y_{..j})^2 - BTSS_1 - BTSS_2$	$(g - 1) \times (n - 1)$
Split-plot residual	$RSS_2 = TSS_2 - ISS_2 - BTSS_2 - TSS_1$	$(m - g) \times (n - 1)$
Split-plot total	$TSS_2 = \sum_{h=1}^g \sum_{i=1}^{m_h} \sum_{j=1}^n (y_{hij} - y_{...})^2$	$nm - 1$

F_1 : To test for difference in expected response between treatments:

$$F_1 = \{BTSS_1 / (g - 1)\} / \{RSS_1 / (m - g)\},$$

F_2 : To test for lack of parallel group mean profile:

$$F_2 = \{ISS_2 / [(g - 1)(n - 1)]\} / \{RSS_2 / [(m - g)(n - 1)]\},$$

ANOVA with Repeated Measures (Cont'd)

(Example) The Sitka spruce 1989

ANOVA table

Source of variation	Sum of squares	df
Between treatments	$BTSS_1 = n \sum_{h=1}^g m_h (y_{h..} - y_{...})^2$	$g - 1$
Whole plot residual	$RSS_1 = TSS_1 - BTSS_1$	$m - g$
Whole plot total	$TSS_1 = n \sum_{h=1}^g \sum_{i=1}^{m_h} (y_{hi.} - y_{...})^2$	$m - g$
Between times	$BTSS_2 = m \sum_{j=1}^n (y_{..j} - y_{...})^2$	$n - 1$
Treatment by time interaction	$ISS_2 = \sum_{j=1}^n \sum_{h=1}^g m_h (y_{h.j} - y_{..j})^2 - BTSS_1 - BTSS_2$	$(g - 1) \times (n - 1)$
Split-plot residual	$RSS_2 = TSS_2 - ISS_2 - BTSS_2 - TSS_1$	$(m - g) \times (n - 1)$
Split-plot total	$TSS_2 = \sum_{h=1}^g \sum_{i=1}^{m_h} \sum_{j=1}^n (y_{hij} - y_{...})^2$	$nm - 1$

Under the null hypotheses, $\alpha_h = 0$, $\tau_j = 0$, and $\lambda_{hj} = 0$.

v^2 can be 0, which means no covariances between repeated measurements for each subject.

ANOVA with Repeated Measures (Cont'd)

(Example) The Sitka spruce 1989

Results: The following table presents a split-plot ANOVA for the 1989 sitka spruce data. We ignore possible chamber effects to ease comparison with the results from the robust analysis.

Source of variation	Sum of squares	df	Mean square
Between treatments	17.106	1	17.106
Whole plot residuals	249.108	77	3.235
Whole plot total	266.214	78	3.413
Between times	41.521	7	5.932
Treatment by time interaction	0.045	7	0.0064
Split-plot residual	5.091	539	0.0094
Split-plot total	312.871	631	

What are the test statistics for F_1 , F_2 ?

ANOVA with Repeated Measures (Cont'd)

Assumptions in the Sitka Spruce Example

$$U_{hi} \sim IN(0, \nu^2), \quad \varepsilon_{hij} \sim IN(0, \sigma^2)$$



$$\text{cov}(y_{hij}, y_{h'i'j'}) = \begin{cases} \sigma^2 + \nu^2 & \text{if } i = i', j = j' & \text{Homogeneity of variances} \\ \nu^2 & \text{if } i = i', j \neq j' & \text{Sphericity} \\ 0 & \text{if } i \neq i', j \neq j' & \text{Independence} \end{cases}$$

ANOVA with Repeated Measures (Cont'd)

Assumptions in the Sitka Spruce Example

Between subjects

- When analysing data from different participants, you have to assume homogeneity of variances, (for example) same variances in all groups.
- Homogeneity of variances is tested with Levene's test.

Within subjects

- Since your subjects are always the same, there will necessarily be a relation between the measurements in the various conditions in a within subj design. While the relations cannot be independent they should be the same between all pairs.

$$\text{variance}_{A-B} \approx \text{variance}_{A-C} \approx \text{variance}_{B-C}$$

ANOVA with Repeated Measures (Cont'd)

Sphericity

The validity of (some of the) F-tests rests on assumption of the split-plot model - in particular, type H matrix including compound symmetry (the equal correlation).

Type H matrix

$$\begin{pmatrix} \sigma_1 & \frac{\sigma_1 + \sigma_2}{2} - \lambda & \frac{\sigma_1 + \sigma_3}{2} - \lambda & \frac{\sigma_1 + \sigma_4}{2} - \lambda \\ & \sigma_2 & \frac{\sigma_2 + \sigma_3}{2} - \lambda & \frac{\sigma_2 + \sigma_4}{2} - \lambda \\ & & \sigma_3 & \frac{\sigma_3 + \sigma_4}{2} - \lambda \\ & & & \sigma_4 \end{pmatrix}$$

Sphericity assume that variances in the differences between conditions is equal (basically it means that the correlation between treatment levels is the same).

Sphericity test: perform a statistical test for the structure, type H matrix (Type H matrix satisfies sphericity).

ANOVA with Repeated Measures (Cont'd)

Sphericity

Start	1 Month	2 Months	Start – 1 Month	Start – 2 Months	1 Month – 2Months
64	62	74	2	-10	-12
63	63	62	0	1	1
66	65	71	1	-6	-6
106	100	84	6	22	16
67	66	66	1	1	0
120	112	107	8	13	5
62	63	61	-1	1	2
72	71	48	1	24	23
83	73	65	10	18	8
77	65	62	12	15	3
Variance			21.33	140.67	100.89

ANOVA with Repeated Measures (Cont'd)

Mauchly Sphericity Test

Likelihood ratio test:

H_0 : type H matrix is preserved vs H_1 : not H_0

$$-\log \left[\frac{\sup_{H_0} L(w)}{\sup_{H_0 \cup H_1} L(\Omega)} \right] \sim \chi^2(df)$$

When the significance level of the Mauchly's test is < 0.05 , then sphericity cannot be assumed

ANOVA with Repeated Measures (Cont'd)

Remarks

- Repeated measures ANOVA is a somewhat ad hoc approach to analysis.
- For each model we need to find appropriate sums of squares, F-test.
- If sphericity is violated,
 - Use modified tests (Huynh-Feldt or Greenhouse-eisser).
 - Use MANOVA
- The method assumes balanced data (the same number and timing of measurements across subject). F-test validity is not guaranteed for unbalanced data.

ANOVA with Repeated Measures (Cont'd)

Limitations

- ANOVA/MANOVA assume categorical predictors.
- They assume everyone is measured at the same time (time is categorical) and at equally spaced time intervals.
- You don't get parameter estimates (just p-values).
- Missing data must be imputed.
- They require restrictive assumptions about the correlation structure.